

Customer Churn & Retention Analysis

Business Intelligence & Customer Analytics Project

Business Problem

Customer retention is one of the most important factors affecting long-term business growth and recurring revenue. High churn rates can significantly impact profitability, customer lifetime value, and operational stability.

This project analyzes customer churn behavior to identify patterns related to customer abandonment, engagement, contract structure, and revenue impact. The goal is to provide actionable insights that support customer retention strategies and business decision-making.

Project Objectives

- Analyze customer churn behavior and retention trends.
- Identify high-risk customer segments.
- Measure revenue impact caused by churn.
- Evaluate the relationship between engagement and retention.
- Create a business intelligence dashboard for executive decision-making.
- Generate strategic recommendations to improve retention performance.

Tools Used

- SQL
- Python
- Google Sheets

Dataset Overview

The dataset was manually structured to simulate a subscription-based business environment. Each row represents an individual customer profile containing subscription, financial, engagement, and support-related information.

CustomerID: Unique customer identifier

Age: Customer age

Gender: Customer gender

ContractType: Monthly, Quarterly, or Annual subscription
TenureMonths: Number of months as a customer
MonthlyFee: Monthly subscription payment
TotalSpent: Total customer spending
SupportTickets: Number of support requests
PaymentMethod: Customer payment method
LastLoginDays: Days since last platform activity
CustomerSegment: Basic, Premium, Enterprise
SatisfactionScore: Customer satisfaction score
Churn: Indicates whether the customer left

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	CustomerID	Age	Gender	ContractType	TenureMonths	MonthlyFee	TotalSpent	SupportTickets	PaymentMethod	LastLoginDays	Churn	Customer Segment	Satisfaction Score
2	C001		22 Male	Monthly	2	15	30	3	Credit Card	40	Yes	Basic	5
3	C002		34 Male	Quarterly	3	32	96	2	PayPal	10	No	Premium	7
4	C003		56 Female	Annual	2	180	360	1	Bank Transfer	1	Yes	Enterprise	10
5	C004		45 Male	Monthly	17	32	544	10	Debit Card	1	Yes	Enterprise	9
6	C005		57 Female	Monthly	29	32	928	11	Debit Card	1	Yes	Enterprise	9
7	C006		37 Female	Monthly	10	32	320	22	PayPal	22	Yes	Enterprise	7
8	C007		30 Female	Monthly	11	32	352	0	PayPal	19	Yes	Basic	3
9	C008		30 Female	Monthly	23	32	736	12	PayPal	0	Yes	Basic	2
10	C009		30 Male	Annual	7	180	1260	0	PayPal	1	No	Basic	10
11	C010		30 Male	Annual	11	180	1980	0	PayPal	1	No	Basic	9
12	C011		30 Male	Quarterly	21	32	672	6	PayPal	1	No	Premium	10
13	C012		30 Male	Quarterly	11	32	352	8	PayPal	1	No	Premium	5
14	C013		30 Female	Annual	12	180	2160	0	PayPal	1	No	Premium	5
15	C014		30 Male	Quarterly	23	32	736	11	PayPal	60	Yes	Premium	2
16	C015		30 male	Quarterly	33	32	1056	12	PayPal	30	No	Enterprise	4
17	C016		30 male	Quarterly	4	32	128	2	PayPal	22	No	Premium	3
18	C017		30 Female	Monthly	22	15	330	3	PayPal	33	No	Basic	4
19	C018		30 Female	Monthly	10	15	150	9	PayPal	0	No	Premium	3
20	C019		30 Female	Monthly	1	15	15	5	PayPal	1	No	Enterprise	5
21	C020		30 Female	Monthly	10	15	150	12	PayPal	1	No	Enterprise	5
22	C021		30 Male	Annual	7	180	1260	2	PayPal	1	No	Enterprise	4
23	C022		30 Male	Annual	26	180	4680	2	PayPal	1	No	Enterprise	4
24	C023		30 Male	Quarterly	8	32	256	2	PayPal	1	Yes	Basic	8
25	C024		30 Male	Quarterly	8	32	256	2	PayPal	1	Yes	Basic	9
26	C025		30 Female	Annual	12	180	2160	2	PayPal	1	Yes	Basic	9
27													

Key Performance Indicators (KPIs)

Churn Rate

Measures the percentage of customers who abandoned the service.

$$\text{Churn Rate} = \frac{\text{Customers Lost}}{\text{Total Customers}} \times 100$$

Retention Rate

Measures the percentage of customers retained.

$$\text{Retention Rate} = 100 - \text{Churn Rate}$$

Customer Lifetime Value (CLV)

Represents estimated customer value over time.

$$CLV = \text{Average Revenue Per Customer} \times \text{Customer Lifespan}$$

Revenue Lost

Measures monthly recurring revenue lost due to churned customers.

SQL Analysis

Churn by Contract Type

```
SELECT
  ContractType,
  COUNT(*) AS customers,
  SUM(CASE WHEN Churn = 'Yes' THEN 1 ELSE 0 END) AS
  churned_customers
FROM customers
GROUP BY ContractType;
```

Insight

Monthly subscriptions presented the highest churn rate, significantly outperforming annual contracts. This indicates that long-term contracts improve customer retention and reduce abandonment risk.

Revenue Lost by Churn

```
SELECT
  SUM(MonthlyFee) AS lost_revenue
FROM customers
WHERE Churn = 'Yes';
```

Insight

Enterprise customer churn generated disproportionate revenue loss despite representing a smaller customer segment. Retaining high-value customers is critical for revenue stability.

Churn by Customer Segment

```
SELECT
  CustomerSegment,
  COUNT(*) AS customers,
  SUM(CASE WHEN Churn='Yes' THEN 1 ELSE 0 END) AS churned
FROM customers
GROUP BY CustomerSegment;
```

Insight

Basic customers showed the highest churn rate, while Premium customers demonstrated significantly stronger retention performance.

Python Analysis

Python was used for:

- data cleaning,
- validation,
- exploratory analysis,
- and visualization.

The analysis focused on identifying relationships between churn behavior, customer engagement, subscription type, and financial performance.

Key Areas Explored

- Contract distribution
- Customer segmentation
- Support activity
- Revenue impact
- Engagement behavior
- Satisfaction trends

Dashboard Design Process

The dashboard structure was initially designed in Figma before being implemented in Power BI. This process helped improve visual hierarchy, KPI organization, and user experience.

Dashboard Design Goals

- Executive-level readability
- Clear KPI visibility
- Minimalist visual structure
- Efficient storytelling
- Easy navigation and filtering

Dashboard Overview

The Power BI dashboard was designed to provide a comprehensive overview of customer retention performance and churn risk indicators.

Dashboard Components

KPI Cards

- Total Customers
- Churn Rate
- Retention Rate
- Revenue Lost

Visualizations

- Churn by Contract Type
- Revenue by Customer Segment
- Support Tickets vs Churn
- Customer Distribution
- Engagement Analysis

Key Insights

- Monthly subscription customers presented the highest churn probability.
- Annual contracts significantly improved customer retention.
- Enterprise customer churn generated major revenue impact.
- Basic-tier customers demonstrated weaker retention performance.
- Low platform engagement correlated strongly with customer abandonment.
- Support ticket activity showed a relationship with churn risk.

11. Business Recommendations

Improve Long-Term Contract Adoption

Encourage customers to migrate from monthly plans to annual subscriptions through discounts and retention incentives.

Strengthen Basic Customer Retention

Develop personalized engagement campaigns targeting Basic-tier customers with high churn risk.

Prioritize Enterprise Retention

Enterprise customers generate higher recurring revenue and should receive proactive support and retention strategies

Monitor Customer Engagement

Track inactivity indicators such as login frequency to identify potential churn risks earlier.

Improve Customer Experience Monitoring

Review and validate satisfaction measurement methods to improve predictive accuracy.

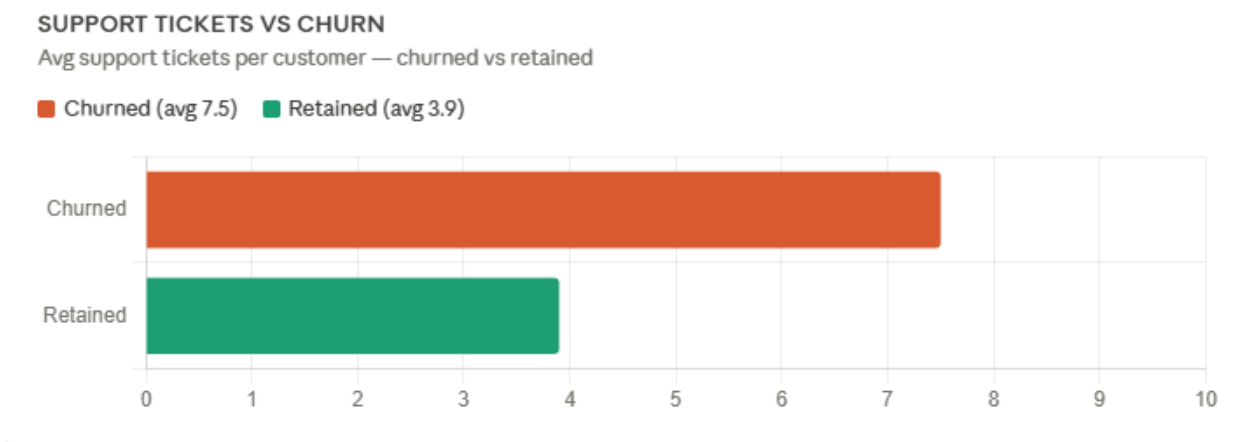
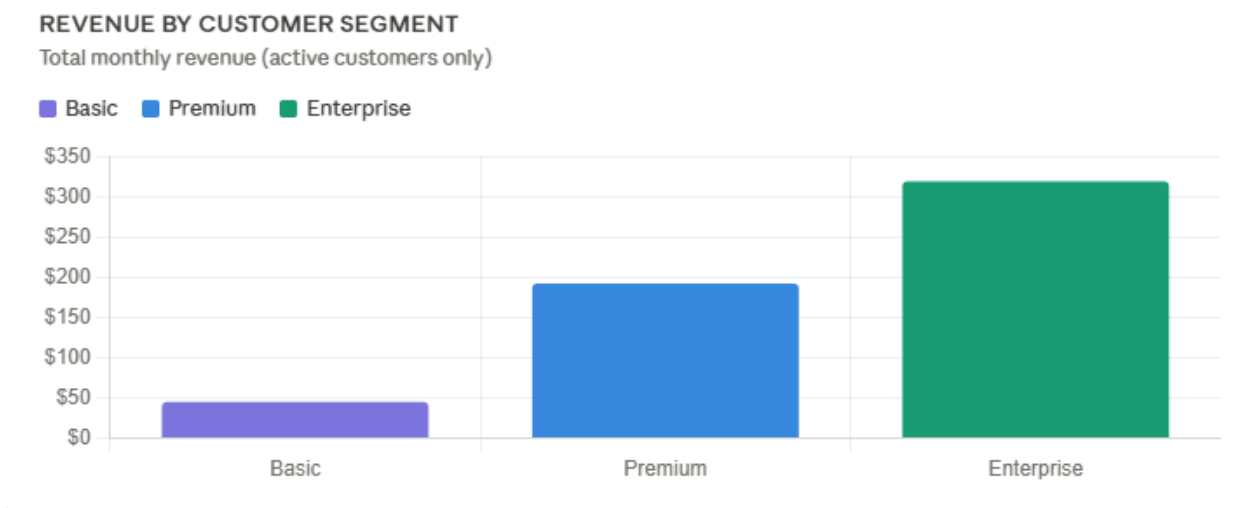
12. Conclusion

This project demonstrates how customer behavior, subscription structure, and engagement patterns directly impact churn and revenue performance.

By combining SQL, Python, Power BI, and dashboard design principles, the analysis provides actionable business insights that support strategic retention initiatives and executive decision-making.

The project also highlights the importance of combining technical analytics skills with business understanding and visual storytelling to create effective business intelligence solutions.

DASHBOARDS.



CUSTOMER DISTRIBUTION

By segment

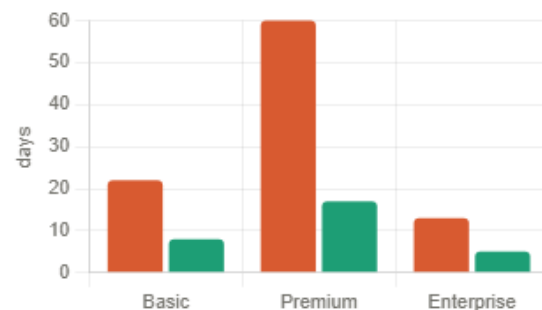
Basic 36% Premium 28% Enterprise 36%



ENGAGEMENT ANALYSIS

Avg days since last login

Churned Retained



Churn by contract type — Monthly contracts have the highest churn rate at 60% (6 out of 10 customers lost), nearly double the retained count. Annual contracts are the healthiest, losing only 2 out of 7 customers (28.6%). The pattern is clear: longer commitment correlates directly with lower churn. Customers on monthly plans have the lowest switching cost and act on it.

Revenue by segment — Enterprise accounts for 58% of active monthly revenue (\$319/month), making it the most valuable segment by far. However, it also carries a 44% churn rate, which creates real concentration risk — losing one or two Enterprise customers has an outsized impact on revenue compared to losing several Basic ones. High value and high volatility in the same bucket is a dangerous combination.

Support tickets — Churned customers averaged 7.5 support tickets vs. 3.9 for retained customers — nearly twice as many. This is a strong leading indicator: elevated ticket volume signals friction well before a customer actually leaves. Building an early-warning system around ticket spikes could give the team a meaningful window to intervene before churn happens.

Customer distribution — Basic and Enterprise each have 9 customers, so they look balanced on paper. But Basic churns at 66.7% while Premium churns at just 14.3% — a 4× difference. Premium is clearly the most stable segment despite being the smallest. This suggests Premium customers find enough value to stay, and points to upsell from Basic to Premium as a viable retention lever.

Engagement — The one Premium customer who churned had gone 60 days without logging in, far above the dataset average. That pattern looks like passive abandonment — the product simply faded out of their routine — rather than active dissatisfaction. It's a different problem than high-ticket churn, and it calls for a different response: re-engagement campaigns and usage nudges rather than customer success outreach.

Key conclusions

The data tells a consistent story across every dimension analyzed: churn in this customer base is not random — it follows predictable patterns that point to specific, actionable levers.

Contract structure is the single strongest retention driver. The jump from Monthly (60% churn) to Annual (28.6% churn) is the largest single variable gap in the entire dataset. Incentivizing customers to commit to longer contracts — through discounts, feature unlocks, or onboarding perks — is likely the highest-ROI intervention available right now.

Enterprise is both the greatest asset and the greatest risk. It generates 58% of active revenue but loses nearly half its customers. A single Enterprise churn can wipe out the equivalent of multiple Basic customers gained. This segment needs dedicated retention investment: proactive account management, regular check-ins, and clear escalation paths before issues become exits.

Support tickets are an early warning signal. With churned customers averaging nearly twice the ticket volume of retained ones, the support queue is effectively a churn heatmap. Flagging customers who cross a ticket threshold — say, 6 or more open tickets — and routing them to a retention workflow could catch at-risk customers before they decide to leave.

Basic is leaking customers at an unsustainable rate. Two out of every three Basic customers churn. At that rate, the segment is a revolving door rather than a growth foundation. Either the product isn't delivering enough value at that tier, or these customers were never a strong fit to begin with. The fix is either improving the Basic experience or accelerating the path to Premium, where retention is dramatically better.

Passive disengagement is a separate problem from active dissatisfaction.

The Premium churn case — 60 days without a login — suggests some customers don't leave angry, they just drift away. That requires a different playbook: automated re-engagement triggers, in-product nudges, and usage milestone reminders, rather than reactive customer success calls.

Bottom line: the highest-priority actions are converting Monthly customers to Annual contracts, protecting the Enterprise base with dedicated account management, and building a ticket-based early warning system. Together, these three moves address the majority of churn risk visible in this dataset.